

ECO 362 Financial Investments

Precept 9: Term Structure Model / Derivatives

November 17 – November 21, 2025

1 Term Structure Model (Continued)

2 Futures / Forwards

3 Interest Rate Swap

Extensions: Inflation risk

- Let $y_b(t)$ and $y_i(t)$ denote nominal and real yields of horizon t , respectively
 - Real yield means yield on inflation-indexed treasuries (aka TIPS)
- Let $i(t)$ denote average (continuously-compounded) inflation over horizon t :

$$i(t) \equiv \frac{1}{t} \log \left(\frac{CPI_t}{CPI_0} \right)$$

- Then we can show

$$\underbrace{y_b(t) - y_i(t)}_{\text{break-even inflation}} = \underbrace{\frac{1}{t} \log \left(\mathbb{E} \left[e^{ti(t)} \right] \right)}_{\text{expected inflation}} + \underbrace{IRP_t}_{\text{inflation risk premium}}$$

IRP depends on the correlation between inflation $i(t)$ and the SDF

- If inflation $i(t)$ correlates positively with SDF, nominal is risky and inflation risk premium is positive

Extensions: Credit Risk

- Let $y_b(t)$ denote nominal risk-free yield as before
- Let $y_d(t)$ denote (nominal) risky yield on a defaultable bond (e.g. a corporate bond) of horizon t
- Let $d(t)$ denote the recovery rate in the event of default of the risky t -horizon bond
- Then we can show

$$\underbrace{y_d(t) - y_b(t)}_{\text{credit spread}} = \underbrace{-\frac{1}{t} \log(\mathbb{E}[d(t)])}_{\text{expected default}} + \underbrace{CRP_t}_{\text{credit risk premium}}$$

CRP depends on the correlation between recovery rate $d(t)$ and the SDF

- If recovery $d(t)$ correlates negatively with SDF (that is, the default risk $1 - d(t)$ correlates positively with SDF), bond is extra-risky (i.e. beyond just expected default) and credit spread increases

1 Term Structure Model (Continued)

2 **Futures / Forwards**

3 Interest Rate Swap

- A **future/forwards contract** is agreement to buy Q units of an asset at predetermined price F in T periods from now
 - Futures: traded on exchange. Forwards: traded OTC.
 - In principle: zero payment for entering the contract today. In reality: margin requirements (e.g. 5% of notional).
- Terms:
 - F : Futures/forward price (paid T periods from now)
 - T : Maturity
 - S_0 : Spot price (current price of the asset)
 - S_T : Price at maturity
 - $S_0 \times Q$: Notional amount
 - Long: buying futures/forwards contracts, payoff $S_T - F$
 - Short: selling futures/forwards contracts, payoff $F - S_T$

Futures / Forwards: no-arbitrage pricing

- For simplicity, assume that the underlying asset does not pay dividends or interest, and continuous compounding.
- Trading strategy with 0 payment today:
 - Borrow S_0 at risk-free rate r_b
 - Buy one unit of the asset at price S_0
 - Short (sell) futures at futures price F and maturity T .
- Payoff at maturity.
 - Repay the loan and interest: $-S_0 e^{r_b T}$
 - Price of the asset at T : S_T
 - Net payoff from the futures contract: $F - S_T$
- Under no-arbitrage, the payoff at maturity should be 0:

$$F = S_0 e^{r_b T}$$

Return of futures

- What is the return of futures if I just buy and hold till maturity?
- Long strategy:
 - Long futures at price F and maturity T
 - Lend $Fe^{-r_b T}$ with risk-free rate r_b
- Initial investment: $Fe^{-r_b T}$
- Payoff at maturity: $S_T - F + Fe^{-r_b T} \cdot e^{r_b T} = S_T$
- Assume continuous compounding, then the return is

$$r_p = \frac{1}{T} \log \left[\frac{S_T}{Fe^{-r_b T}} \right] = r_b + \frac{1}{T} \log \left(\frac{S_T}{S_0} \right) + \frac{1}{T} \log \left(\frac{S_0}{F} \right)$$

- Expected excess return:

$$\mathbb{E}[r_p - r_b] = \frac{1}{T} \mathbb{E} \left[\log \left(\frac{S_T}{S_0} \right) \right] + \frac{1}{T} \log \left(\frac{S_0}{F} \right)$$

1 Term Structure Model (Continued)

2 Futures / Forwards

3 Interest Rate Swap

- A contract to exchange a floating short-term interest rate $r_{b,t}$ for a fixed rate c for T periods
- Let N be the notional amount
- Payoff is realized in each period $t = 1, \dots, T$
 - Investor that receives floating and pays fixed: $(r_{b,t} - c)N$
 - Investor that receives fixed and pays floating: $(c - r_{b,t})N$
- c is determined so that the swap contract is fair to both sides, that is, the value of the swap is 0

- Strategy 1: trade in risk-free bonds
 - At $t = 0$, borrow \$1 at risk-free rate $r_{b,1}$ and uses it to buy 1 unit a T period bond with coupon rate c at price P_c ;
 - At $t = 1, \dots, T - 1$, receives c , repay the debt $1 + r_{b,1}$ and borrow \$1 at rate $r_{b,t+1}$;
 - At $t = T$, receives $1 + c$ and repay the debt $1 + r_{b,T}$
 - Net payoff in each period $t = 1, \dots, T$ is $c - r_{b,t}$
- Strategy 2: enter swap contract with $N=\$1$ by receiving fixed and paying floating
- Strategy 1 and 2 have the same payoffs in $t = 1, \dots, T$. By no arbitrage, they should have the same payoff in $t = 0$, that is,

$$P_c = 1$$

- $P_c = 1$ implies that the coupon bond with rate c is traded at par
- That is, c should be the **par yield** with maturity T

$$c = \frac{1 - P_b(T)}{\sum_{t=1}^T P_b(t)}$$

where $P_b(t)$ is the price of t -year zero-coupon bond with \$1 face value.